A DIAGONAL LATTICE RECTANGLE TOUCHING ALL FOUR SIDES OF A 655×104 BOX: A COUNTEREXAMPLE TO KAGEY'S CONJECTURE ON THE SMALLEST ENTRIES OF OEIS A338885

MATH0

Abstract. For $n \geq 2$ let $\mathcal{R}(n)$ be the set of positive integers k such that some rectangle with integer vertices and no side parallel to the x-axis lies in the box $[0, n] \times [0, k]$ and touches all four of its sides; these sets are the rows of the OEIS triangle A338885. In 2020 P. Kagey conjectured that $\min \mathcal{R}(n) = \mathrm{A228286}(n)$, where $\mathrm{A228286}(n) = \min\{x + yz : x, y, z \geq 1,\ xy + z = n\}$. We show that the conjecture is false: the rectangle with vertices $(4, 0)$, $(655, 42)$, $(651, 104)$, $(0, 62)$ shows $104 \in \mathcal{R}(655)$, while every value $x + yz$ with $xy + z = 655$ is at least 105 (in fact $\mathrm{A228286}(655) = 115$). To our knowledge the counterexample is new; it is the result of a short computation, and the conjecture is still listed as open in the OEIS. The refutation is checked in Lean 4 against a statement that math0 formalized from the OEIS entry, whose definitions are checked by the kernel against the listed data; this statement is not part of the formal-conjectures repository.

1. Introduction

A *diagonal lattice rectangle* is a rectangle whose vertices lie in $\mathbb{Z}^2$ and none of whose sides is parallel to the x-axis. The OEIS entry A338885, contributed by Peter Kagey on November 14, 2020 [7], is the irregular triangle whose n-th row ($n \geq 2$) lists, in increasing order, all k such that a diagonal lattice rectangle touches all four sides of an $n \times k$ rectangle. The rows begin

$$\mathcal{R}(2) = \{2\}, \quad \mathcal{R}(3) = \{3\}, \quad \mathcal{R}(4) = \{4, 5\}, \quad \mathcal{R}(5) = \{4, 5, 7\}, \quad \mathcal{R}(6) = \{6, 9, 10\}, \ \ldots$$

The row lengths form A338886 [8], and the entry links Kagey's table of the first 100 rows and a Code Golf Stack Exchange challenge on the same question [1]. The entry contains two conjectures, about the smallest and the largest number in each row. The first one compares the row minimum with the sequence A228286 [6], contributed by Andy Niedermaier on August 19, 2013:

$$\mathrm{A228286}(n) = \min\{x + yz : x, y, z \in \mathbb{Z}_{\geq 1},\ xy + z = n\} \qquad (n \geq 2).$$

According to a comment in that entry, one term of A228286 was the subject of a problem in the Team Round of the 2013 ARML contest.

In this note we show that the first conjecture is false (Theorem 3). By computation, the smallest counterexample is $n = 655$, where $\min \mathcal{R}(655) = 104$ and $\mathrm{A228286}(655) = 115$. The reason is visible in the parametrization of Section 3.1: the elements of $\mathcal{R}(n)$ are the numbers $pr + qs$ with $pq + rs = n$ and $p, q, r, s \geq 1$, and $\mathrm{A228286}(n)$ is the minimum over the quadruples with an entry equal to 1. The element 104 of $\mathcal{R}(655)$ comes from $655 = 2 \cdot 2 + 21 \cdot 31$, that is, $(p, q, r, s) = (2, 2, 21, 31)$.

Status and prior work. The conjecture is stated without a counterexample in the current revision #22 of A338885 (September 11, 2021); the math0 referees compared this revision with the local mirror of the OEIS data and found them identical. The entry's table covers only rows 2 to 101, all below the first counterexample. The referees also checked the cross-referenced

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entries A228286, A228287, A228288, A338886, A338887, A033638, A085582 and A113751, and the answers and comments of the Code Golf question [1]; none of them mentions a resolution. Web searches (arXiv, MathOverflow, Mathematics Stack Exchange, the SeqFan mailing list, the OEIS wiki) found no proof or counterexample. The conjecture is not among the OEIS statements of the formal-conjectures repository [2]. We therefore classify the result as *new*. The caveat is that the counterexample is found by a straightforward search, so others may have observed it without recording it.

The second conjecture of the entry, that $\max \mathcal{R}(n)$ equals A033638$(n) = \lfloor n^2/4 \rfloor + 1$ [5], is true and elementary (Remark 9); it is not the subject of this note.

2. Statement

Throughout, the $n \times k$ *box* is $B_{n,k} = [0, n] \times [0, k] \subset \mathbb{R}^2$.

Definition 1. For integers $n \geq 2$ and $k \geq 0$, write $k \in \mathcal{R}(n)$ if there are $P_0 = (x_0, y_0) \in \mathbb{Z}^2$ and $u = (u_1, u_2)$, $v = (v_1, v_2) \in \mathbb{Z}^2$ such that

(1) $u \cdot v = u_1 v_1 + u_2 v_2 = 0$ and $u_2 \neq 0$, $v_2 \neq 0$;
(2) the four points P_0, $P_0 + u$, $P_0 + u + v$, $P_0 + v$ lie in $B_{n,k}$;
(3) each of the lines $x = 0$, $x = n$, $y = 0$, $y = k$ contains one of these four points.

Every rectangle with integer vertices is $\{P_0 + su + tv : s, t \in [0, 1]\}$ for a corner $P_0 \in \mathbb{Z}^2$ and nonzero perpendicular side vectors $u, v \in \mathbb{Z}^2$, and its sides are parallel to u or to v. So $u_2, v_2 \neq 0$ says exactly that no side is parallel to the x-axis; it also forces $u, v \neq 0$, so the rectangle is non-degenerate. Squares are allowed; for example $\mathcal{R}(2) = \{2\}$ comes from the square with vertices $(1, 0), (2, 1), (1, 2), (0, 1)$. A convex polygon lies in the box if and only if its vertices do, and if it lies in the box, it meets the side on the line $x = 0$ if and only if the minimum of the linear function $(x, y) \mapsto x$ over the polygon is 0. This minimum is attained at a vertex, and the same holds for the other three sides. So (2) and (3) say exactly that the rectangle lies in the $n \times k$ box and touches all four of its sides, and $\mathcal{R}(n)$ is the n-th row of A338885.

The comment section of A338885 contains, verbatim:

> `Conjecture: The smallest number in the n-th row is A228286(n).`

The entry, including this comment, is by Peter Kagey, November 14, 2020.

Conjecture 2 (Kagey)**.** *For every $n \geq 2$,* A228286(n) *is the least element of* $\mathcal{R}(n)$.

Theorem 3. $104 \in \mathcal{R}(655)$ *and* A228286$(655) \geq 105$. *In particular, Conjecture 2 is false.*

In fact A228286$(655) = 115$ and $\min \mathcal{R}(655) = 104$, and 655 is the smallest counterexample (Proposition 8); these facts are computations that are not formalized.

Lean formalization. The statement was formalized by math0 in `lean/Math0/Statements/OEIS/A338885.lean`. It was reviewed for fidelity and frozen (SHA-256 `c2422329...a45e84db`) before any proof attempt. The definitions on which the statement depends, the set `A228286Set` used in the proof, and the statement itself are, verbatim except that docstrings are omitted (the computational definitions `inRowB`, `row`, `flatAux` and `a`, used only for the data check, are described below):

```
namespace Math0.Statements.A338885

def InBox (n k : ℕ) (p : ℤ × ℤ) : Prop :=
  0 ≤ p.1 ∧ p.1 ≤ n ∧ 0 ≤ p.2 ∧ p.2 ≤ k

def vertices (x₀ y₀ u₁ u₂ v₁ v₂ : ℤ) : List (ℤ × ℤ) :=
  [(x₀, y₀), (x₀ + u₁, y₀ + u₂), (x₀ + u₁ + v₁, y₀ + u₂ + v₂), (x₀ + v₁, y₀ + v₂)]
```

```
def InRow (n k : ℕ) : Prop :=
  ∃ x₀ y₀ u₁ u₂ v₁ v₂ : ℤ,
    -- the sides `u` and `v` are perpendicular (so the parallelogram is a rectangle)
    u₁ * v₁ + u₂ * v₂ = 0 ∧
    -- no side is parallel to the x-axis
    u₂ ≠ 0 ∧ v₂ ≠ 0 ∧
    -- the rectangle lies in the box
    (∀ p ∈ vertices x₀ y₀ u₁ u₂ v₁ v₂, InBox n k p) ∧
    -- it touches the sides `x = 0`, `x = n`, `y = 0`, `y = k`
    (∃ p ∈ vertices x₀ y₀ u₁ u₂ v₁ v₂, p.1 = 0) ∧
    (∃ p ∈ vertices x₀ y₀ u₁ u₂ v₁ v₂, p.1 = n) ∧
    (∃ p ∈ vertices x₀ y₀ u₁ u₂ v₁ v₂, p.2 = 0) ∧
    (∃ p ∈ vertices x₀ y₀ u₁ u₂ v₁ v₂, p.2 = k)

def A228286Set (n : ℕ) : Set ℕ :=
  {m | ∃ x y z : ℕ, 0 < x ∧ 0 < y ∧ 0 < z ∧ x * y + z = n ∧ m = x + y * z}

def A228286List (n : ℕ) : List ℕ :=
  (List.range' 1 (n - 1)).flatMap fun x => (List.range' 1 (n - 1)).filterMap fun y =>
    if x * y < n then some (x + y * (n - x * y)) else none

def A228286 (n : ℕ) : ℕ := (A228286List n).min?.getD 0

def statement : Prop :=
  ∀ n : ℕ, 2 ≤ n → IsLleast {k : ℕ | InRow n k} (A228286 n)
```

Here `IsLleast S m` means $m \in S$ and $m \le s$ for all $s \in S$. The file also proves the following facts (proofs omitted, data lists abbreviated). In it, `row n` is the list of all $k \le \lfloor n^2/4 \rfloor + 1$ that pass a Boolean test for the existence of $0 < a < n$ and $0 < c < k$ with $a(n-a) = c(k-c)$, and `a` is the flattened triangle read from these lists.

```
theorem inRow_iff_param (n k : ℕ) :
    InRow n k ↔ ∃ a b c d : ℕ, 0 < a ∧ 0 < b ∧ 0 < c ∧ 0 < d ∧
      a + b = n ∧ c + d = k ∧ a * b = c * d
theorem le_of_inRow {n k : ℕ} (h : InRow n k) : k ≤ n * n / 4 + 1
theorem mem_row (n k : ℕ) : k ∈ row n ↔ InRow n k
theorem row_sorted (n : ℕ) : (row n).Pairwise (· < ·)
theorem a_data : (List.range 71).map (fun i => a (i + 2)) =
    [2, 3, 4, 5, 4, 5, 7, 6, 9, 10, 5, 7, 8, 11, 13, ..., 14, 15, 17, 20, 22, 23]
theorem mem_A228286List (n m : ℕ) : m ∈ A228286List n ↔ m ∈ A228286Set n
theorem isLeast_A228286 {n : ℕ} (hn : 2 ≤ n) : IsLeast (A228286Set n) (A228286 n)
theorem A228286_data : (List.range 70).map (fun i => A228286 (i + 2)) =
    [2, 3, 4, 4, 6, 5, 7, 6, 6, 7, 9, 7, ..., 16, 16, 18, 17, 23, 21, 25, 17]
```

Encoding. `InRow n k` is Definition 1 with $P_0 = (x_0, y_0)$; the list `vertices` is $P_0, P_0+u, P_0+u+v, P_0+v$, and all coordinates are integers, so no truncated subtraction or division occurs in it. `A228286Set n` is the set $\{x + yz : x, y, z \ge 1,\ xy + z = n\}$ from the OEIS definition. `A228286 n` is the minimum of an explicit finite list, and `isLeast_A228286` proves that for $n \ge 2$ it is the least element of `A228286Set n`; the truncated subtraction `n - x * y` in `A228286List` is guarded by `x * y < n`. Both sequences have offset 2, and `statement` quantifies over all $n \ge 2$ with no further hypothesis.

Data check. The theorems `a_data` and `A228286_data` are proved by `decide +kernel`. They check all 71 listed terms of A338885 (rows 2 to 12 and the listed beginning of row 13) and all 70 listed terms of A228286, A228286(2), ..., A228286(71). By `mem_row` and `row_sorted`, the list `row n` is the increasing enumeration of the set of all k with `InRow n k`, so the first check

compares the geometric definition, not a formula, with the OEIS data. The referees compared both lists with the entries by script, and one referee compared the rows computed from the parametrization of `inRow_iff_param` with Kagey's table of the first 100 rows; all agree.

3. Proof

The proof has two parts: an explicit rectangle, and a finite search that bounds A228286(655) from below. It does not compute A228286(655): an attempt to evaluate `A228286 655`, whose list ranges over $654^2 \approx 4.3 \cdot 10^5$ pairs (x, y), with `decide +kernel` did not finish (the process was killed after about seven minutes, probably for lack of memory).

Lemma 4 (`inRow_655_104`)**.** $104 \in \mathcal{R}(655)$.

Proof. Take $P_0 = (4, 0)$, $u = (651, 42)$ and $v = (-4, 62)$. Then $u \cdot v = 651 \cdot (-4) + 42 \cdot 62 = -2604 + 2604 = 0$, and $u_2 = 42 \neq 0$, $v_2 = 62 \neq 0$. The four vertices are

$$P_0 = (4, 0), \quad P_0 + u = (655, 42), \quad P_0 + u + v = (651, 104), \quad P_0 + v = (0, 62),$$

all in $[0, 655] \times [0, 104]$. The lines $x = 0$, $x = 655$, $y = 0$, $y = 104$ contain $(0, 62)$, $(655, 42)$, $(4, 0)$, $(651, 104)$, respectively. □

In Lean, the proof supplies $(x_0, y_0, u_1, u_2, v_1, v_2) = (4, 0, 651, 42, -4, 62)$ and the four touching vertices, and `simp` and `norm_num` check the conditions.

Lemma 5 (`search`)**.** *For all integers* $1 \leq x, y \leq 104$ *with* $xy < 655$*,*

$$x + y\,(655 - xy) \geq 105.$$

Proof. This is a finite check of the 2391 pairs (x, y) in the range; the smallest value of $x+y(655-xy)$ among them is 115. In Lean the statement is

```
private theorem search : ∀ x < 105, ∀ y < 105, 0 < x → 0 < y → x * y < 655 →
    105 ≤ x + y * (655 - x * y) := by
  decide +kernel
```

The bounded quantifiers, implications and inequalities have decidability instances from the core library. With `decide +kernel`, type-checking the proof requires the kernel itself to reduce the resulting Boolean to `true`, which it does by evaluating all $105 \cdot 105 = 11025$ cases $x, y \in \{0, \ldots, 104\}$ (those with $x = 0$, $y = 0$ or $xy \geq 655$ hold because a hypothesis fails), using its arbitrary-precision arithmetic on natural-number literals. Unlike `native_decide`, no compiled code is trusted. □

Lemma 6 (`ge_of_mem`)**.** *If* $x, y, z \geq 1$ *and* $xy + z = 655$*, then* $x + yz \geq 105$*.*

Proof. Suppose $m = x + yz \leq 104$. Then $x \leq m \leq 104$, and since $z \geq 1$, $y \leq yz \leq m \leq 104$. Moreover $xy = 655 - z < 655$ and $z = 655 - xy$. Lemma 5 gives $m = x + y(655 - xy) \geq 105$, a contradiction. □

In Lean the hypothesis is `m ∈ A228286Set 655`; the arithmetic steps are done by `Nat.le_mul_of_pos_right` and `omega`.

Proof of Theorem 3. The first claim is Lemma 4. For the second, A228286(655) $\in \{x + yz : x, y, z \geq 1,\ xy + z = 655\}$ by `isLeast_A228286` (with $n = 655 \geq 2$), so A228286(655) ≥ 105 by Lemma 6. Now suppose the conjecture held. Then for $n = 655$, A228286(655) is a lower bound of $\mathcal{R}(655)$, and $104 \in \mathcal{R}(655)$ gives A228286(655) ≤ 104, a contradiction. The Lean proof is, verbatim:

```
theorem result : ¬ statement := by
  intro h
  have hle : A228286 655 ≤ 104 := (h 655 (by norm_num)).2 inRow_655_104
  have hge : 105 ≤ A228286 655 := ge_of_mem (isLeast_A228286 (by norm_num)).1
  omega
```

Here `(h 655 _).2` is the lower-bound half of `IsLeast`; the membership half is not used. □

3.1. **Further remarks (not formalized).** The following facts explain the counterexample. Apart from the parametrization of `inRow_iff_param` and the bound `le_of_inRow`, they are not part of the Lean development.

Lemma 7. *For $n \geq 2$,*

$$\mathcal{R}(n) = \{c+d : a,b,c,d \geq 1,\ a+b=n,\ ab=cd\} = \{pr+qs : p,q,r,s \geq 1,\ pq+rs=n\}.$$

Proof. The first equality is `inRow_iff_param`. Given a rectangle as in Definition 1, put $a=|u_1|$, $b=|v_1|$, $c=|u_2|$, $d=|v_2|$. The x-coordinates of the vertices range over an interval of length $|u_1|+|v_1|$, and by (2) and (3) this interval is $[0,n]$; so $a+b=n$, and likewise $c+d=k$. From $u_1v_1 = -u_2v_2 \neq 0$ we get $a,b \geq 1$ and $ab=cd$. Conversely, given a,b,c,d, the corner $P_0=(b,0)$ with $u=(a,c)$ and $v=(-b,d)$ has vertices $(b,0)$, (n,c), (a,k), $(0,d)$ and $u\cdot v = cd-ab = 0$. Lemma 4 is this construction with $(a,b,c,d)=(651,4,42,62)$.

For the second equality, let $ab=cd$, put $p=\gcd(a,c)$ and write $a=pq$, $c=pr$ with $\gcd(q,r)=1$. Then $qb=rd$, so $r \mid b$; writing $b=rs$ gives $d=qs$. Hence $n=pq+rs$ and $k=pr+qs$. Conversely $(a,b,c,d)=(pq,rs,pr,qs)$ satisfies $ab=cd$. □

The representation $(p,q,r,s) \mapsto (pq+rs,\ pr+qs)$ is invariant under $(p,q,r,s)\mapsto(q,p,s,r)$ and $(p,q,r,s)\mapsto(s,r,q,p)$, which together move any entry to any position. With $s=1$, $(p,q,r)=(y,x,z)$ gives $pq+rs=xy+z$ and $pr+qs=x+yz$. Hence

$$\{x+yz : x,y,z\geq 1,\ xy+z=n\} = \{pr+qs : pq+rs=n,\ \min(p,q,r,s)=1\} \subseteq \mathcal{R}(n),$$

so $\mathrm{A228286}(n) \in \mathcal{R}(n)$ and $\min\mathcal{R}(n) \leq \mathrm{A228286}(n)$ for every $n\geq 2$. Conjecture 2 is therefore equivalent to the claim that, for every n, the minimum of $pr+qs$ subject to $pq+rs=n$ is attained at a quadruple containing a 1. For $n=655$, the identity $655 = 2\cdot 2 + 21\cdot 31$ gives $(p,q,r,s)=(2,2,21,31)$ and $pr+qs=42+62=104$.

Proposition 8 (computation)**.** $\mathrm{A228286}(655) = 115$*, attained for example at* $(x,y,z)=(6,109,1)$ *and* $(31,21,4)$*. The row* $\mathcal{R}(655)$ *begins* $104,115,121,125,135,137$*. Among* $2\leq n\leq 2000$*,* $\min\mathcal{R}(n) \neq \mathrm{A228286}(n)$ *holds exactly for the following nine values of* n*:*

n	655	845	909	1019	1229	1435	1439	1759	1931
$\min\mathcal{R}(n)$	104	116	149	128	140	160	152	168	176
$\mathrm{A228286}(n)$	115	130	153	131	149	161	181	173	197

We computed these values by exhaustive search in Python, using Lemma 7 for $\min\mathcal{R}(n)$. They agree with the referees' independent computations, which found the same counterexamples up to $n=1500$; the value $\mathrm{A228286}(655)=115$ also agrees with Kagey's table of A228286 [6]. None of this is formally verified, and Theorem 3 does not depend on it.

Remark 9 (The largest entry)**.** The second conjecture of A338885 states that the largest number in row n is $\mathrm{A033638}(n) = \lfloor n^2/4\rfloor + 1$. By Lemma 7, if $k=c+d\in\mathcal{R}(n)$ then $c+d \leq cd+1 = ab+1 \leq \lfloor n^2/4\rfloor+1$, because $(c-1)(d-1)\geq 0$ and $ab\leq (a+b)^2/4$; this upper bound is the theorem `le_of_inRow` of the statement file. It is attained by $(a,b,c,d)=(\lfloor n/2\rfloor, \lceil n/2\rceil, 1, \lfloor n/2\rfloor\lceil n/2\rceil)$. Only the upper bound is formalized; we make no claim about the novelty of this observation.

4. Machine verification

Files and declarations. The proof is in the file `lean/Math0/Frontier/OEIS/A338885.lean` of the math0 repository, `https://github.com/Foundation-0/math0`. It imports the frozen statement file `lean/Math0/Statements/OEIS/A338885.lean` (Section 2) and the audit commands in `lean/Math0/Audit.lean`. The main declaration is

`Math0.Frontier.A338885.result : ¬ Math0.Statements.A338885.statement`.

The auxiliary declarations `search`, `ge_of_mem` and `inRow_655_104` are private; they are Lemmas 5, 6 and 4. The only proof by `decide +kernel` in the proof file is the search of Lemma 5; the data checks of the statement file are not used by the proof.

Audit. The file ends with

```
#math0_settles Math0.Frontier.A338885.result Math0.Statements.A338885.statement
```

This command is defined in `lean/Math0/Audit.lean` for statements that math0 formalizes itself. It fails unless (1) `statement` is a definition of type `Prop` without universe parameters; (2) `statement` and `result` depend on no axioms other than `propext`, `Classical.choice` and `Quot.sound`, which rejects `sorryAx`, the axioms `Lean.ofReduceBool` and `Lean.trustCompiler` introduced by `native_decide`, and custom axioms; and (3) the type of `result` is syntactically the statement or its negation, after removing metadata annotations and with no unfolding. Running `lake env lean Math0/Frontier/OEIS/A338885.lean` in the directory `lean` reports:

```
math0 audit: 'Math0.Statements.A338885.statement' depends on axioms []
math0 audit: 'Math0.Frontier.A338885.result' depends on axioms [Classical.choice, Quot.sound, propext]
math0 statement check: 'Math0.Frontier.A338885.result' refutes 'Math0.Statements.A338885.statement'
```

Neither file contains `sorry`, `admit`, `native_decide`, `axiom` or `set_option`.

Toolchain and reproduction. We use Lean 4 [4] `v4.33.1` and Mathlib [3] `v4.33.1`, pinned in `lean/`. The repository also pins formal-conjectures [2] at commit `c77fdd4`, which this result does not use. To reproduce, run `scripts/verify.sh` from the repository root; it fetches the Mathlib build cache, builds all math0 results with `lake build` (which re-runs every audit command), and checks that the pinned dependencies are unmodified. To check this result alone, run `lake build Math0.Statements.OEIS.A338885` (including the data checks, about 45 seconds) and then `lake env lean Math0/Frontier/OEIS/A338885.lean` in the directory `lean`.

References

[1] P. Kagey, *Rectangles in rectangles*, Code Golf Stack Exchange, question 213754, October 2020, `https://codegolf.stackexchange.com/q/213754`.

[2] M. Firsching, P. Lezeau, S. Mercuri, M. Z. Horváth, Y. Dillies, C. Sönne, E. Wieser, F. Zhang, T. Hubert, B. Agüera y Arcas, and P. Kohli, *Formal Conjectures: An Open and Evolving Benchmark for Verified Discovery in Mathematics*, arXiv:2605.13171, 2026. Repository: `https://github.com/google-deepmind/formal-conjectures`.

[3] The mathlib Community, *The Lean mathematical library*, in: Proceedings of the 9th ACM SIGPLAN International Conference on Certified Programs and Proofs (CPP 2020), ACM, 2020, 367–381.

[4] L. de Moura and S. Ullrich, *The Lean 4 theorem prover and programming language*, in: Automated Deduction – CADE 28, Lecture Notes in Comput. Sci. **12699**, Springer, 2021, 625–635.

[5] The OEIS Foundation Inc., `https://oeis.org/A033638`, Quarter-squares plus 1.

[6] The OEIS Foundation Inc., `https://oeis.org/A228286`, entry by A. Niedermaier (2013).

[7] The OEIS Foundation Inc., `https://oeis.org/A338885`, entry by P. Kagey (2020).

[8] The OEIS Foundation Inc., `https://oeis.org/A338886`, entry by P. Kagey (2020).